Jodh Singh Nahal

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Education

University of British Columbia *Vancouver, BC*
Bachelor of Applied Science in Computer Engineering *Expected May 2028*
- Presidential Scholars Award ($60,000), Dean's Honour List, available 4 or 8 months, CGPA: 84.7%.

Technical Skills

Languages: Python, C/C++, Shell, Perl, SQL
Reliability & Performance: Code Profiling, Regression Testing, CI/CD, Multi-threading, Linux/Unix, MQTT
Tools: Docker, Jenkins, Git, Jira, Confluence, GDB

Experience

Hardware Verification Engineering Intern *Vancouver, BC*
AMD *Sept 2026 – Present*
- Building automation scripts and regression workflows to improve validation efficiency and coverage closure.
- Debugging RTL behavior through simulation and waveform analysis.
- Contributing to verification of the IOHUB subsystem for next-generation AMD processor architectures.
- Verifying IOMMU L1 cache behavior using UVM in a constrained-random testbench.

Software Test / Embedded Development Co-op *Burnaby, BC*
Xantrex *May 2024 – Aug 2025*
- Achieved 7× device capacity, 90% lower CPU usage, and ∼2 s lower latency after the C++ migration.
- Reduced build times by 5× by implementing CI/CD cross-compilation pipelines with Docker, Jenkins, and CMake.
- Troubleshot CAN/BLE communication faults on embedded inverter and charger platforms using oscilloscopes, logic analyzers, PCAN-USB adapters, and BLE sniffers.
- Built Python Hardware-in-the-Loop (HIL) CAN and UART simulators to run 100+ verification tests on the Raspberry Pi gateway, validating edge cases and stress-testing real-time performance.
- Performed code reviews and system-level regression testing, reporting issues on Jira and Confluence.
- Developed an object-oriented, multi-threaded, non-blocking C++ firmware architecture on Linux-based Raspberry Pi gateways to parse RV-C CAN frames from STM32 devices and publish structured messages over MQTT.
- Replaced a ∼60k-line Python implementation with the C++ architecture.

Firmware / Embedded Systems Developer *Vancouver, BC*
UBC Rover *Sept 2023 – July 2026*
- Automated firmware validation with GitHub Actions CI and Catch2 unit tests to improve regression coverage.
- Led a team of 3 engineers building a distributed embedded control system across ESP32 (FreeRTOS), STM32, Arduino, and TI LaunchPad MCUs over SPI/I2C.
- Aggregated inputs from switches, buttons, potentiometers, encoders, and joysticks and streamed data over UART.

Undergraduate Research Assistant *Vancouver, BC*
UBC Electrical and Computer Engineering — Supervisor: Dr. Lutz Lampe *May 2026 – Sept 2026*
- Modeling nonlinear SSHI interface circuits and verifying simulated voltage waveforms against theoretical models.
- Optimizing power extraction of piezoelectric energy harvesting interface circuits (SEH, SSHI, SECE) through LTspice and Simulink/Simscape simulation.
- Researching battery-less platforms for the Internet of Underwater Things via simulation of piezoelectric interface circuits.

Projects

OS161 Kernel Development | *C, GDB, MIPS, Operating Systems*
- Engineered thread synchronization primitives, including blocking locks and condition variables, to resolve race conditions in kernel space.
- Implemented system calls including fork, execv, and file I/O.

Memory Allocator Implementation | *C, Pointers, Systems Programming*
- Developed a custom malloc, free, and realloc using an implicit free list with 8-byte alignment and block headers.